1. What is Multiple Linear Regression?

Multiple Linear Regression models the relationship between a dependent variable yyy (what we want to predict) and **several independent variables** (features, predictors):

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \varepsilon$$

- y: response (dependent variable)
- $x_1, x_2, \ldots, x_p$: predictors (independent variables)
- β_0: intercept (value of y when all predictors = 0)
- $\beta_1, \beta_2, \ldots, \beta_p$: coefficients (effect of each predictor on y)
- ε: error term (random noise not explained by predictors)

Intuition Behind the Model

- In **simple linear regression** (one predictor), we fit a straight line.
- In **multiple regression**, instead of a line, we fit a **hyperplane** in higher dimensions.
    - For 2 predictors: the model is a plane in 3D.
    - For p predictors: the model is a p-dimensional hyperplane.

Each β_j tells us **how much y changes if x_j increases by 1 unit, holding all other predictors constant.**

Matrix Form

To simplify, we write MLR in **matrix notation**:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Where:

- $\mathbf{y}$: $n \times 1$ vector of outcomes
- $\mathbf{X}$: $n \times (p+1)$ matrix of predictors (first column all 1's for intercept)
- $\boldsymbol{\beta}$: $(p+1) \times 1$ vector of coefficients
- $\boldsymbol{\varepsilon}$: $n \times 1$ vector of errors

Example:

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1p} \\ 1 & x_{21} & x_{22} & \dots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \dots & x_{np} \end{bmatrix}$$

4. How Do We Find $\boldsymbol{\beta}$? (Least Squares)

We choose $\boldsymbol{\beta}$ so the predicted values $\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta}$ are **as close as possible** to the actual $\mathbf{y}$.

We minimize the **sum of squared residuals (SSR)**:

$$\text{SSR} = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$$

Taking derivative and setting to zero gives the **normal equations**:

$$\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}$$

If $\mathbf{X}^\top \mathbf{X}$ is invertible:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

This is the **closed-form solution**.

i	x_1	x_2	y
1	1	2	5
2	2	1	7
3	3	3	8
4	4	5	12
5	5	4	11
6	6	7	17
7	7	6	14
8	8	8	20

We write the model in matrix form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ \vdots & \vdots & \vdots \\ 1 & x_{81} & x_{82} \end{bmatrix}.$$

For our dataset the design matrix X (8×3) is

$$\mathbf{X} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 3 & 3 \\ 1 & 4 & 5 \\ 1 & 5 & 4 \\ 1 & 6 & 7 \\ 1 & 7 & 6 \\ 1 & 8 & 8 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 5 \\ 7 \\ 8 \\ 12 \\ 11 \\ 17 \\ 14 \\ 20 \end{bmatrix}.$$

We solve the normal equations

$$\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top y.$$

Compute $X^\top X$ and $X^\top y$ (numbers):

$$X^\top X = \begin{bmatrix} 8 & 36 & 36 \\ 36 & 204 & 201 \\ 36 & 201 & 204 \end{bmatrix}, \qquad X^\top y = \begin{bmatrix} 94 \\ 506 \\ 508 \end{bmatrix}.$$

The inverse $(X^\top X)^{-1}$ is

$$(X^\top X)^{-1} = \begin{bmatrix} 0.625 & -0.05555556 & -0.05555556 \\ -0.05555556 & 0.17283951 & -0.16049383 \\ -0.05555556 & -0.16049383 & 0.17283951 \end{bmatrix}.$$

Now multiply to get $\hat{\beta}$:

$$\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top y = \begin{bmatrix} 2.416666666666587 \\ 0.7037037037037112 \\ 1.370370370370376 \end{bmatrix}.$$

It's nice to see these as exact rational numbers:

$$\hat{\beta}_0 = \tfrac{29}{12} = 2.416666\ldots, \quad \hat{\beta}_1 = \tfrac{19}{27} = 0.703703\ldots, \quad \hat{\beta}_2 = \tfrac{37}{27} = 1.370370\ldots.$$

So the fitted regression equation is

$$\boxed{\hat{y} = 2.4166667 \;+\; 0.7037037\, x_1 \;+\; 1.3703704\, x_2}.$$

Take observation 4: $x_1 = 4,\ x_2 = 5$. Plug into the formula:

$$\hat{y} = 2.4166667 + 0.7037037 \cdot 4 + 1.3703704 \cdot 5.$$

Compute step by step (digit-by-digit arithmetic):

1. $0.7037037 \times 4 = 2.8148148.$
   (Because $0.7037037 \times 2 = 1.4074074$; times 4 doubles that → 2.8148148.)
2. $1.3703704 \times 5 = 6.8518520.$
   (Because $1.3703704 \times 10 = 13.703704$, half of that is 6.851852.)
3. Sum them:

$$2.4166667 + 2.8148148 = 5.2314815$$

then

$$5.2314815 + 6.8518520 = 12.0833335.$$

Rounded to a sensible number of digits, $\hat{y} \approx 12.0833333$.

Compare to actual $y = 12$. The residual for this observation is

$$e_4 = y_4 - \hat{y}_4 = 12 - 12.0833333 \approx -0.0833333,$$

which matches the residuals computed earlier.

Overall goodness-of-fit numbers (from the same computation)

- Sum of squared errors (SSE) $= \sum (y_i - \hat{y}_i)^2 \approx 8.6111111.$
- Total sum of squares (SST) $\approx 183.5.$
- $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} \approx 0.95307$ — the model explains about 95.3% of variance in y for this dataset.